

Charotar University of Science and Technology
M.Sc. (Advanced Organic Chemistry) Semester II Examination (Regular)
OC 709: Organic Chemistry-II

Date: 04/05/2011

Time: 10.00 a.m. to 1.00 p.m.

Total Marks: 70

Instructions to the candidates:

1. Provide all necessary information as required in the answer sheet.
2. Attempt all questions.
3. Questions in **Part A** should be answered in the question paper itself. Please **do not write** your candidate ID number on the question paper for Part A.
4. Write answers for Section I and Section II of **Part B** in separate answer sheets.
5. Use of cell phones is strictly prohibited.
6. Use of non-programmable calculators may be allowed if needed.

PART A

Total marks: 20

Choose the correct option and put $\sqrt{}$ mark in front of it:

1. Name of the 4-membered cyclic intermediate in Wittig reaction is
(a) oxaphosphetane (b) betain
(c) triphenyl phosphine oxide (d) phosphorous ylide
2. Robinson annulation is the reaction of α , β -unsaturated ketone with
(a) α -carbanion of cyclic ketone (b) alkene
(c) α -carbocation of cyclic ketone (d) α -halo ketone
3. α - Alkylation and α - acetylation of ketones can be done by,
(a) Robinson annulation (b) Wittig reaction
(c) Stork enamine reaction (c) None of the above
4. Wittig reaction fails for synthesis of
(a) disubstituted olefins (b) trisubstituted olefins
(c) tetrasubstituted olefins (d) none of these
5. According to Dakin reaction oxidation of carbonyl group present in ortho or para position of phenolic group is possible using _____.
(a) alkaline H_2O_2 (b) acidic H_2O_2 (c) peracid (d) HgO
6. Jones reagent is
(a) chromium trioxide-acetate sulphuric acid (b) chromium trioxide-pyridine complex
(c) Pyridinium chlorochromate (d) pyridinium dichromate
7. In case of oxidation of carbonyl compound with peracid, the correct migrating aptitude is
(a) $3^\circ\text{-alkyl} > 2^\circ\text{-alkyl} > 1^\circ\text{-alkyl} > \text{methyl}$ (b) $3^\circ\text{-alkyl} > 2^\circ\text{-alkyl} > \text{methyl} > 1^\circ\text{-alkyl}$
(c) $3^\circ\text{-alkyl} > 1^\circ\text{-alkyl} > 2^\circ\text{-alkyl} > \text{methyl}$ (d) $\text{methyl} > 1^\circ\text{-alkyl} > 2^\circ\text{-alkyl} > 3^\circ\text{-alkyl}$
8. In presence of osmium tetroxide and $NaHSO_3$, *cis*-maleic acid oxidises to
(a) *trans*-tartaric acid (b) *meso*-tartaric acid
(c) *cis*- tartaric acid (d) mixture of *cis* and *trans*- tartaric acid
9. Toluene on reduction with Na-liq. NH_3 gives
(a) 2, 4-dihydro product (b) 1, 4-dihydro product
(c) 2, 5-dihydro product (d) 1, 5-dihydro product

10. _____ fails with acid sensitive compounds as well as high molecular weight compounds
- (a) Clemmenson reduction (b) Wolf Kishner reduction
(c) Both (a) and (b) (d) Birch reduction
11. Which transition metal is used as a catalyst in hydrogenolysis?
- (a) copper (b) nickel (c) chromium (d) none of these
12. NaBH_4 cannot be used for the preparation of
- (a) 1° alcohols (b) 2° alcohols
(c) 3° alcohols (d) epoxide
13. In protic solvents, Bamford Stevens reaction involves formation of _____, whereas in aprotic solvents it involves formation of _____.
- (a) carbanion, carbene (b) carbene, carbanion
(c) carbocation, carbene (d) carbene, carbocation
14. Hydroboration takes place with _____ addition and follows _____ rule.
- (a) anti, Markownikoff's rule (b) syn, Markownikoff's rule
(c) anti, anti-Markownikoff's rule (d) syn, anti-Markownikoff's rule
15. self-condensation of aliphatic nitriles catalyzed by base to form enamine is termed as
- a) Stobbe condensation b) Prins reaction
c) Thorpe reaction d) Shapiro reaction
16. The _____ catalyzed C-C coupling between aryl halides or vinyl halides and activated alkene in the presence of base is referred as _____.
- (a) palladium, heck reaction (b) MgCl_2 , Peterson olefination
(c) acid, Prins reaction (d) NaOH , Sommet-Houser rearrangement
17. For an alkyl halide, the ease of formation of Grignard reagent is of the order
- (a) $\text{RCI} > \text{RBr} > \text{RI}$ (b) $\text{RI} > \text{RBr} > \text{RCI}$ (c) $\text{RBr} > \text{RI} > \text{RCI}$ (d) $\text{RCI} > \text{RI} > \text{RBr}$
18. DCC is a powerful _____.
- (a) oxidizing agent (b) reducing agent
(c) dehydrating agent (d) oxidizing and reducing agent
19. The alkane is synthesized from two different alkyl halides with the help of
- (a) Grignard reagent (b) Gilman reagent
(c) DCC (d) LDA
20. In acid solution, Mn(VII) is reduced to
- (a) Mn(II) (b) Mn(III) (c) Mn(IV) (d) Mn(V)

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Part-B

Marks: 50

SECTION-I

Ques 1:

- (I) Discuss Robinson annulation with mechanism and state its disadvantages. (4)
- (II) Give mechanism for Horner-Wodsworth-Emmons modification. (3)

OR

- (II) Explain stork enamine reaction with mechanism and give any two applications. (3)
- (III) How nature of the ylide and carbonyl compound affect the formation of E and Z alkene in Wittig reaction? (3)

OR

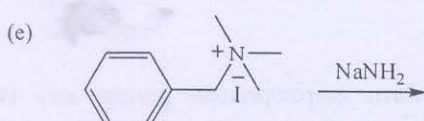
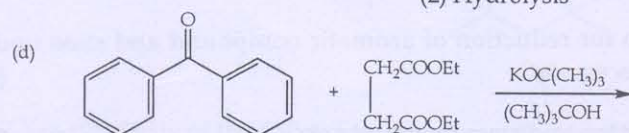
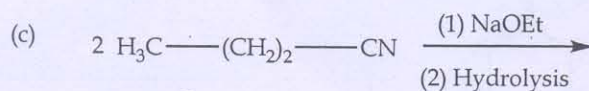
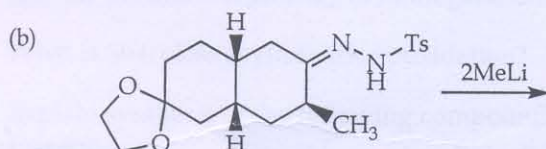
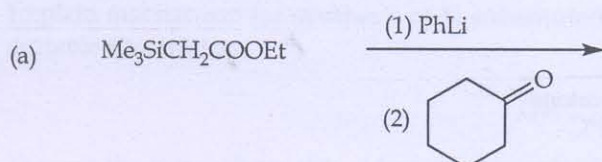
- (III) How complications arising in Robinson annulation can be overcome (3)

Ques 2:

- (I) Discuss Prins reaction with mechanism. Give any two of its applications. (4)
- (II) Briefly Explain mechanism of Stobbe Condensation. (3)

OR

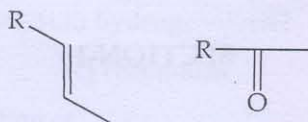
- (II) Explain mechanism for Hydroboration of alkene using diborane. (3)
- (III) Complete the following reactions (Any 3) (3)



SECTION-II

Ques 3:

- (I) Discuss mechanism of oxidation for the following compounds using selenium dioxide as the oxidizing agent. Also give two reactions showing oxidation of alkynes with SeO_2 . (4)

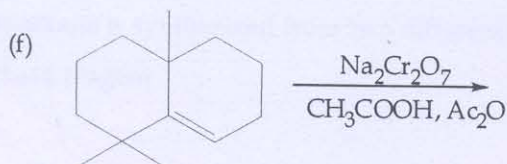
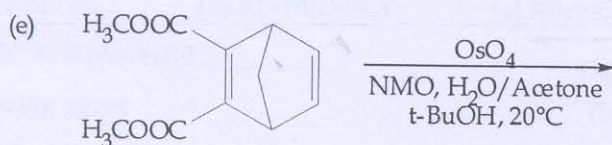
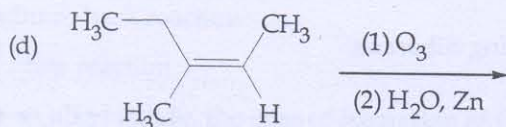
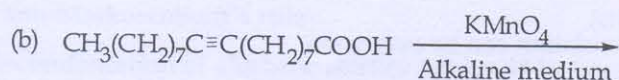
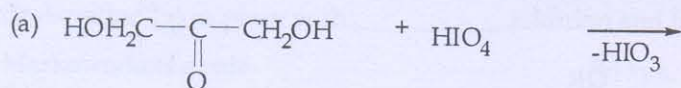


- (II) Give preparation of NBS. Explain mechanism for oxidation of 2° alcohol and α -hydroxy carboxylic acid with NBS. (3)

OR

- (II) Explain mechanism for oxidation of monohydric alcohols as well as *cis* and *trans* 1,2-diols with lead tetra-acetate. (3)

- (III) Complete the following reactions giving product only (Any three). (3)



Quest 4:

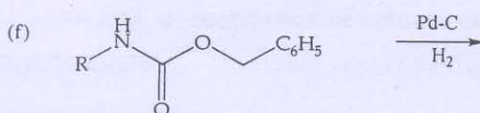
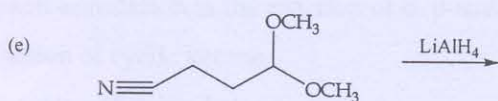
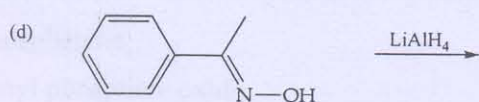
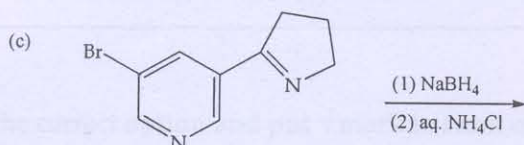
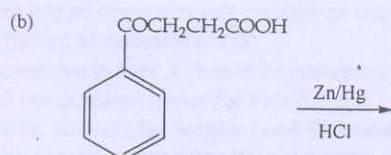
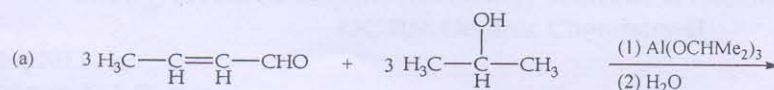
- (I) What is Birch reduction? Give mechanism for reduction of aromatic compound and state under which condition the proton donor is not necessary. (4)
- (II) Discuss mechanism of Wolf Kishner reduction and state its drawback. (3)

OR

- (II) Explain reduction of carbonyl compounds with aluminium isopropoxide giving any two applications. (3)

(III) Complete the following reactions (Any 3).

(3)



Quest 5:

(I) Using Grignard reagent, give preparation and mechanism of tertiary butyl alcohol from ester and isopropyl alcohol from aldehyde. (4)

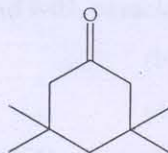
(II) Explain mechanism for synthesis of N-substituted amide from acid and amine using DCC giving suitable illustration. (3)

OR

(II) Discuss the stereochemistry of homogeneous hydrogenation using Wilkinson's catalyst. (3)

(III) What is Sharpless asymmetric epoxidation? (1)

(IV) Explain synthesis of the following compound from α, β -unsaturated ketone using Gilman reagent. (2)



OR

(IV) Give synthesis of the following diketone using acyl halide and 1,3-dithiane. (2)

